

Validation Benchmarks (Schäfer-Turek physical campaign)

Naming convention: V = Validation. The order follows the standard Schäfer-Turek physical cases first: 2D-1, 2D-2, 2D-3. Refinement and diagnostic tests come afterwards.

Common baseline	Value / convention
Discretization	Taylor-Hood P2/P1, theta-method with theta = 0.6
Default mesh and time step	mesh = L0, dt = 0.001 unless the benchmark is a mesh/time-step sweep
Linear/nonlinear setup	SIMPLE preconditioner, SIMPLE alpha = 0.3, Oseen linearization, nonlinear_iterations = 2
Physical constants	rho = 1, D = reference_length = 0.1, H = 0.41, initial condition = rest
Reynolds number	Re = U_ref D / nu. For 2D-1: Um = 0.3, U_ref = 0.2, nu = 1e-3. For 2D-2/2D-3: Um = 1.5, U_ref = 1, nu = 1e-3.
Pressure drop	DeltaP = p_front - p_back, with p_front = (0.15, 0.20), p_back = (0.25, 0.20) in 2D
Output policy	CSV output always enabled. Solution VTU/PVD output only for selected visualization runs or with a sparse output frequency.

ID	Previous ID	Benchmark	Purpose	Physical setup	Sweep	Reference	Main reported quantities
V0	B0	Sanity run	Check pipeline, output folders, CSV files and summary generation.	Short 2D run: mesh L0, Re = 100, dt = 0.001, T = 0.5-1, SIMPLE.	-	No physical reference	Run stability, generated files, no NaN/Inf, stdout/stderr.
V1	B1	2D-1 steady validation	Validate the steady low-Reynolds case.	Schaefer-Turek 2D-1: Re = 20, Um = 0.3, U_ref = 0.2, nu = 1e-3, constant parabolic inlet, T = 10.	-	Schaefer-Turek 2D-1	Cd(T), Cl(T), DeltaP(T), divL2, GMRES mean, runtime.
V2	B3	2D-2 periodic validation	Validate vortex shedding and Strouhal number.	Schaefer-Turek 2D-2: Re = 100, Um = 1.5, U_ref = 1, nu = 1e-3, constant parabolic inlet, T = 20-30.	-	Schaefer-Turek 2D-2	Cd_max, Cl _max, DeltaP over periodic window, St, lift frequency/period. Discard initial transient.
V3	B2	2D-3 ramp validation	Validate the main ramp benchmark.	Schaefer-Turek 2D-3: Re(t) = 0 to 100, Um = 1.5, nu = 1e-3, inlet multiplied by sin(pi t / 8), T = 8.	-	Schaefer-Turek 2D-3	Cd_max, Cl _max, DeltaP(8), divL2 mean/max, runtime.
V4	B4	Mesh refinement on 2D-3	Assess spatial convergence of the 2D-3 ramp case.	Same as V3, fixed dt = 0.001, SIMPLE.	mesh = L0, L1	2D-3 reference + mesh convergence	DoFs, cells, Cd_max, Cl _max, DeltaP(8), divL2, runtime.
V5	B5	Timestep refinement on 2D-3	Assess temporal convergence of the 2D-3 ramp case.	Same as V3, fixed mesh = L0, SIMPLE.	dt = 0.01, 0.005, 0.0025, 0.001	2D-3 reference + time convergence	Cd_max, Cl _max, DeltaP(8), GMRES mean, runtime.
V6	B6	Reynolds sweep on 2D-3	Test robustness when viscosity changes.	2D-3 ramp style setup, mesh L0, dt = 0.001, T = 8, SIMPLE.	Re = 20, 100, 200 with nu = 0.005, 0.001, 0.0005	Diagnostic; not a pure ST reference case	Cd_max, Cl _max, DeltaP(8), divL2, stable, failed_step, runtime.
V7	B9	Grad-div sweep on 2D-3	Study effect of grad-div stabilization on force accuracy and incompressibility.	2D-3 ramp, mesh L0, dt = 0.001, T = 8, SIMPLE, grad_div = true.	gamma = 0, 0.001, 0.01, 0.05, 0.1	Diagnostic / stabilization study	divL2 mean/max, Cd_max, Cl _max, DeltaP(8), GMRES mean, runtime.
V8	B7	Optional 3D validation	Qualitative 3D functionality and reference comparison.	3D ramp: mesh 3D_L0, Um = 2.25, Re = 100, nu = 1e-3, reference_span = 0.41, T = 8 if affordable.	Optional; possibly shorter T for functionality only	Schaefer-Turek 3D-3Z, qualitative if coarse/short	Cd_max, Cl _max, Cs_max, DeltaP(8), GMRES mean, runtime, stable.

Note: using V1, V2, V3 in the order 2D-1, 2D-2, 2D-3 is the clearest order for the report, because it follows the standard Schaefer-Turek numbering. In practice, V3 may be run before V2 because the ramp case is shorter and easier to validate.